

New vegetation implementations into AeoliS

We present recent improvements to the process-based aeolian sediment transport and dune morphology model AeoliS (de Vries et al., 2023; Hoonhout & de Vries, 2016), which now incorporates species-specific characteristics of three dominant dune grass species: European beachgrass (*Ammophila arenaria*, native to western Europe and transplanted to the Pacific Northwest coast of the U.S., Australia, South Africa, and other locations), American beachgrass (*A. breviligulata*, native to the east coast and great lakes region of the U.S. and transplanted on the Pacific Northwest coast), and American dune grass (*Leymus mollis*, native to the U.S. Pacific Northwest coast of the U.S.). These advancements were developed using data from a wind tunnel experiment by Zarnetske et al. (2012), which measured the sand capture efficiency of these species under varying tiller densities and wind speeds.

Using this dataset, we introduced a new approach to integrating vegetation into AeoliS based on the morphological characteristics of the species. The improvements include: (1) dynamically determining the effective vegetation height as a function of wind speed and vegetation density, (2) implementing a new method for representing the vegetation grid, (3) enhancing the representation of vegetation effects on shear velocity reduction and sand deposition within vegetated patches, (4) calibrating the coefficient related to shear stress reduction (R_0) in the Okin (2008) shear stress partitioning model for each species, and (5) developing a quadratic relationship to estimate R_0 across a wider range of tiller densities.

For more detailed explanations and results, please refer to this publication:

Laporte-Fauret, Q., Wengrove, M., Ruggiero, P., Hacker, S. D., Cohn, N., Zarnetske, P. L., & Piercy, C. D. (2024). A new approach to account for species-specific sand capture by plants in an aeolian sediment transport and coastal dune building model. *Journal of Geophysical Research: Earth Surface*, 129, e2024JF007867. <https://doi.org/10.1029/2024JF007867>

1. Blade bending

1.1. Explanations

A new approach was developed to improve the representation of vegetation in the AeoliS model by estimating the frontal area of tillers, taking into account species-specific tiller morphology, tiller density, and bending under different wind speeds. First, based on videos recorded during the wind tunnel experimental runs by Zarnetske et al. (2012), tiller morphology was measured under varying wind speeds and tiller densities, enabling the computation of a metric for blade bending or flexibility. The relative blade height (F_{sp}), which represents blade bending or flexibility, was calculated for each species as the percentage difference in blade height with and without wind for various tiller densities. The measured values were then used to establish species-specific relationships between blade height, wind speed, and tiller density. The

adjusted blade height, $h_{blade,sp,adjusted}$ (m), is then computed using the relaxed blade height of the species, $h_{blade,sp}$ (m), and the species-specific empirical equation for relative blade height, F_{sp} , by:

$$h_{blade,sp,adjusted} = h_{blade,sp} \times \left(\frac{F_{sp}}{100} \right) \quad (1)$$

Limits are applied with $h_{blade,sp,adjusted} = h_{blade,sp}$ if there is no wind and $h_{blade,sp,adjusted} = 0$ if F_{sp} is greater than 100, which indicates that the blade is bent over completely.

Then, the frontal area of each tiller for each species, $\lambda_{tiller,sp}$ (m²) is computed by:

$$\lambda_{tiller,sp} = (w_{stem,sp} \times h_{stem,sp}) + ((w_{blade,sp} \times h_{blade,sp,adjusted}) \times \alpha_{sp}) \quad (2)$$

where $w_{stem,sp}$ (m) and $h_{stem,sp}$ (m) are the width and the height of the stem, respectively, $w_{blade,sp}$ is the width of the blade (m) and α_{sp} is the average number of blades per stem. The frontal area of each tiller is projected onto a 1-mm resolution grid as a circular disk centered on the stem. The disk radius (r) is defined as $r = \sqrt{\lambda_{tiller,sp}/\pi}$, ensuring that the disk area corresponds to the computed frontal area. All grid cells within the disk are assigned a value of 1, indicating full vegetation cover.

Finally, the 1-mm resolution grid with the vegetation is upscaled to a resolution suitable for the numerical model by computing the vegetation cover (from 0 to 1) in each upscaled cell. With this new approach, in the shear stress partitioning model developed by Okin (2008, Equation 9), the height of the canopy h (m) is computed from the stem height $h_{stem,sp}$ (m), the adjusted blade height $h_{blade,sp,adjusted}$ (m) and the vegetation cover in each upscaled cell ρ_{veg} as:

$$h = (h_{stem,sp} + h_{blade,sp,adjusted}) \sqrt{\rho_{veg}} \quad (3)$$

1.2. Implementation into AeoliS

1.2.1. New parameters in constant.py

Table 1. New implemented parameters in constant.py for accounting for leaf flexure in the model. Default values correspond to *Ammophila arenaria* under no-wind conditions.

Parameter	Unit	Description	Default value
rhovegadjusted	—	Vegetation cover after flexure equation	Computed

hvegadjusted	[m]	Height of vegetation after flexure equation	0.61
process_vegflex	–	Enable the process of vegetation flexure (True/False)	False
veg_density	[tiller/m ²]	Vegetation density	148
canopy_height	[m]	Height of the canopy	0.61
canopy_height_stretch	[m]	Height of the stretched canopy	0.61
stem_diam	[m]	Diameter of the stem	0.005
stem_height	[m]	Aboveground height of the stem	0.15
leaf_width	[m]	Width of the leaf	0.0048
leaf_thick	[m]	Thickness of the leaf	0.001
leaf_height	[m]	Height of the leaf	0.47
leaf_height_adjusted	[m]	Height of the leaf after flexure equation	0.47
leafToStem	–	Number of leaves per stem	4.2
flexure_a	–	Coefficient <i>a</i> to compute leaf flexure	-4.12
flexure_b	–	Coefficient <i>b</i> to compute leaf flexure	0.02
flexure_c	–	Coefficient <i>c</i> to compute leaf flexure	94.45

1.2.2. New definitions in vegetation.py

The **VegPositionWrite** definition generates plant positions on a high-resolution grid (1 mm) from an initially lower-resolution grid. Based on the input vegetation grid, which specifies the vegetation cover in each cell, the definition calculates the number of tillers per cell according to the frontal area of a tiller for the chosen species, vegetation density, and grid resolution. The output includes the coordinates of plant positions on the high-resolution grid and the number of plants per cell in the input data. The definition is compatible with both 1D and 2D grids.

The **VegFlexure** definition dynamically adjusts vegetation cover (rhovegadjusted) and height (hvegadjusted) based on wind speed and vegetation density. First, it recalculates leaf height (leaf_height_adjusted) by applying a flexure factor derived from wind speed, density, and predefined coefficients (flexure_a, flexure_b, and flexure_c). The adjusted height is constrained to remain within realistic bounds (not exceeding the initial height or dropping below zero). Next,

the definition computes the adjusted vegetation surface and redistributes vegetation cover across the grid, ensuring the cover per cell does not exceed 100%. Finally, it updates vegetation-related outputs, such as adjusted vegetation cover and height. The definition is compatible with both 1D and 2D grids.

2. Deposition lag length

2.1. Explanations

The downwind flow through vegetation depends on various factors, including the initial wind speed, vegetation density, distribution, morphology, and plant porosity. The downwind distance required for the flow to adjust to the presence of vegetation is referred to as the canopy drag length (L_c). Similarly, sand deposition begins after a lag distance, known as the deposition lag length (L_D), which depends on vegetation cover and L_c , as demonstrated by Dickey et al. (2023) through field measurements and modeling in sparsely vegetated dunes.

Initially, the deposition lag length and the canopy lag length were implemented the Raupach et al. (1992) and Okin (2008) shear coupler in AeoliS (Dickey et al. 2023) as:

$$L_D = 0.95L_c - 0.09 \quad (3)$$

$$L_c = \frac{2(1 - \Phi)h_c}{C_D\lambda} \quad (4)$$

where Φ is the fraction of dune grass shoot volume (m^3), C_D is the drag coefficient of each tiller, and λ is the frontal area of grass tillers per unit of bed area (m^2/m^2). These last three parameters are calculated by:

$$\Phi = \alpha N_{tiller} w_{tiller} t_{tiller} \left(\frac{h_{tiller,stretch}}{h_{tiller,relax}} \right) \quad (5)$$

$$C_D = 1 + 10Re^{-\frac{2}{3}} \quad (6)$$

$$\lambda = \alpha N_{tiller} w_{tiller} h_{tiller} \quad (7)$$

where α and N_{tiller} are the average number of blades per stem and the density of tillers in the canopy, respectively, h_{tiller} , w_{tiller} and t_{tiller} represent the height (m), which can be measured from bed to the top of the longest shoot, the width (m), and the thickness (m) of the tillers, respectively. In the equation of the drag coefficient at the tiller scale (Equation 6), the Reynolds number is computed by $Re = u_z w_{tiller} / \nu$ with ν the air viscosity (m^2/s).

In the initial model by Dickey et al. (2023), a mask was applied from the leading edge of vegetation to the deposition lag length (L_D) to remove morphological changes, preventing erosion and deposition in this area and forming a significant deposition gradient once the distance had been exceeded (blue line on Figure 1). We modified this by shifting the shear stress

partitioning model to apply only beyond L_D , allowing sediment mobilization closer to the leading edge and reducing the deposition gradient (orange line on Figure 1).

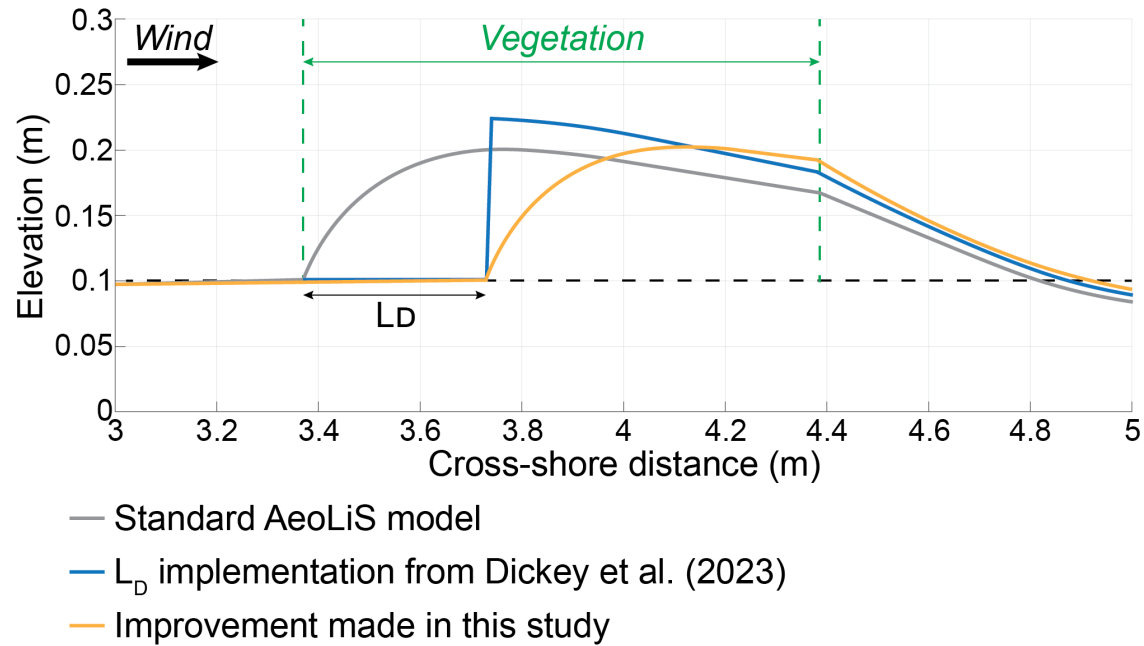


Figure 1. Simulated 1D cross-shore profile of the bed surface changes after 300 min under a constant and unidirectional wind speed of 10 m/s through a 1-m long patch of vegetation fully covered by *Ammophila arenaria*. Vertical green dotted lines delimit the vegetation patch, horizontal black dotted line represents the initial bed surface, and grey, blue, and orange lines represent the model version without L_D , with L_D from Dickey et al. (2023), and the implementation from Laporte-Fauret et al. 2024, respectively.

2.2. Implementation into AeoliS

2.2.1. New parameters in constant.py

Table 2. New implemented parameters in constant.py for accounting for deposition lag length in the model. Default values correspond to *Ammophila arenaria* under no-wind conditions.

Parameter	Unit	Description	Default value
process_veglag	–	Enable the process of vegetation lag length (True/False)	False
veg_density	[tiller/m ²]	Vegetation density	148
canopy_height	[m]	Height of the canopy	0.61

canopy_height_stretch	[m]	Height of the stretched canopy	0.61
stem_diam	[m]	Diameter of the stem	0.005
stem_height	[m]	Aboveground height of the stem	0.15
leaf_width	[m]	Width of the leaf	0.0048
leaf_thick	[m]	Thickness of the leaf	0.001
leaf_height	[m]	Height of the leaf	0.47
leaf_height_adjusted	[m]	Height of the leaf after flexure equation	0.47
leafToStem	–	Number of leaves per stem	4.2

2.2.2. New definitions in vegetation.py

The **adjustment_length** definition calculates the canopy drag length (L_C) and the deposition lag length (L_D) downwind of a vegetated area. It uses vegetation-related parameters (veg_density, canopy_height, canopy_height_stretch, stem_diam, stem_height, leaf_width, leaf_thick, leaf_height, leaf_height_adjusted, leafToStem) and wind characteristics to estimate L_C . An empirical relationship is applied to compute L_D from L_C , with a correction ensuring that which L_D not is negative. The resulting L_D represents the downwind distance required for sand deposition to begin after encountering vegetation

3. Calibration of R0 coefficient in the Okin shear coupler model

3.1. Explanations

In the AeoliS model, the effect of vegetation on shear velocity (u^*) is accounted for through a "shear coupler," which partitions the imposed shear into a component acting on the bare surface (e.g., particle shear as u_s^*) and another on the vegetation (e.g., vegetation as u_{veg}^*), modifying the shear velocity to reflect the influence of roughness elements like vegetation. Two shear couplers are implemented into AeoliS, one coming from the Raupach formulation (Raupach, 1992) and another one coming from the Okin formulation (Okin, 2009).

The Raupach (1992) formulation (Equation 8) uses a dimensionless roughness factor to model vegetation's capacity to reduce wind flow and trap sand but is limited by its assumptions of uniform roughness, lack of downwind sheltering effects, and oversimplification of vegetation's porous nature, making it more appropriate for solid roughness elements than for vegetation. The Raupach formulation is given by:

$$\frac{u_{*,veg}}{u_*} = \frac{1}{\sqrt{1 + \Gamma \rho_{veg}}} \quad (8)$$

where Γ is a roughness parameter set to 16 for dune grass (Durán and Herrmann, 2006).

The Okin (2008) formulation (Equation 9) has an alternative approach in which shear stress reduction is computed using a probability distribution of unvegetated to vegetated area. The Okin (2008) approach (built on experimental observations of Bradley and Mulhearn, 1983) assumes that the shear stress downwind of a porous roughness element is not a fixed value but instead increases gradually downwind using an empirical relation that involves the shear stress, the leeward distance, and the vegetation height. This model contains two coefficients that adjust for the influence of vegetation on both shear stress reduction, R_0 , and on downwind shear stress recovery, C_1 . The Okin formulation is given by:

$$\frac{u_{*,veg}}{u_*}(x) = R_0 + (1 - R_0) \left(1 + e^{-C_1 \frac{x}{h}} \right) \quad (9)$$

where $R_0 = u_{*,veg}/u_*$ for $x = 0$, x is the coordinate relative to the roughness element, C_1 is a factor controlling the shear stress recovery and h is the height of the roughness elements (m), where in AeoliS, h is implemented as the height of the canopy (h_c). Based on the data of Bradley and Mulhearn (1983) using values of sand flow through fences, the best-fit values for the Okin formulation are $R_0 = 0.32$ and $C_1 = 4.8$.

In Laporte-Fauret et al. (2024), to compute the reduction in shear stress depending on plant species, data from a wind tunnel experiment that measured sand capture efficiencies for three plant species under two wind speeds and nine tiller densities were used. First, the initial conditions of the experiment, including wind tunnel dimensions, sand quantity, wind speeds, and the density and morphological characteristics of the tested species, were reproduced in the Aeolis model. In the shear stress partitioning model of by Okin (2008), the shear stress recovery coefficient, C_1 , was set to its default value of 4.8, while R_0 values ranging from 0 to 1 (in increments of 0.01) were systematically tested. Then, the simulated sand capture efficiency was calculated for each tested R_0 value and compared to the observed sand capture efficiency values from the wind tunnel experiment. For each test case, the R_0 value that produced the simulated sand capture efficiency closest to the observed value was selected. To extend the model over a wider range of tiller densities, R_0 coefficient values were interpolated and extrapolated a function of tiller density (d_{veg}), wind speed, and grass species, using a quadratic equation (Equation 10) to describe these relationships for each species and wind speed as:

$$R_0 = a + b \times d_{veg} + c \times d_{veg}^2 \quad (10)$$

where a , b and c are coefficients listed in Table 1.

Table 3. Coefficients of the quadratic equations used for the three dune grass species (i.e., *Ammophila arenaria*, *A. breviligulata* and *Leymus mollis*) for two wind speeds.

Species	Wind speed (m/s)	Coefficient a	Coefficient b	Coefficient c
<i>A. arenaria</i>	9.58	0.98	- 1.08e-4	- 3.86e-7
	5.97	0.99	- 0.57e-4	- 1.13e-7
<i>A. breviligulata</i>	9.58	1.01	- 6.94e-4	6.17e-7
	5.97	0.99	- 3.23e-4	1.94e-7
<i>L. mollis</i>	9.58	1.00	- 8.05e-4	10.68e-7
	5.97	1.01	- 5.29e-4	7.13e-7

3.2. Implementation into AeoliS

3.2.1. New parameters in constant.py

Table 4. New implemented parameters in constant.py for accounting for accounting for variable R_0 Okin coefficient as a function of vegetation density. Default values correspond to *Ammophila arenaria* under no-wind conditions.

Parameter	Unit	Description	Default value
process_okinR0var	–	Enable the process of variable R_0 Okin coefficient depending on vegetation density (True/False)	False
okin_initialred_hws_coef_a	–	Coefficient a for high wind speed to compute R_0 Okin coefficient depending on vegetation density	0.98
okin_initialred_hws_coef_b	–	Coefficient b for high wind speed to compute R_0 Okin coefficient depending on vegetation density	- 1.08e-4
okin_initialred_hws_coef_c	–	Coefficient c for high wind speed to compute R_0 Okin coefficient depending on vegetation density	- 3.86e-7
okin_initialred_lws_coef_a	–	Coefficient a for low wind speed to compute R_0 Okin coefficient depending on vegetation density	0.99
okin_initialred_lws_coef_b	–	Coefficient b for low wind speed to compute R_0 Okin coefficient depending on vegetation density	- 0.57e-4

okin_initialred_lws_coef_c	–	Coefficient c for low wind speed to compute R_0 Okin coefficient depending on vegetation density	- 1.13e-7
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3.2.2. New definitions in vegetation.py

The **vegsh ear** definition runs the shear coupler model to compute shear stress reduction in the lee of plants using either the Raupach (1992) model (implemented in the **compute_raupach_shear** definition for 2D and in the **vegsh ear_raupach** definition for 1D) or the Okin (2008) model (implemented in the **compute_okin_shear** definition for 2D and in the **vegsh ear_okin** definition for 1D). If the Okin (2008) model is selected, the definition collects parameters from the processes **process_vegflex**, **process_okinR0var**, and **process_veglag**, depending on whether each is set to True or False, to compute shear stress reduction in the lee of plants. If the Raupach (1992) model is selected, the definition gathers parameters from the processes **process_vegflex** and **process_veglag**, depending on their activation status (True or False), to compute shear stress reduction.

The **compute_okin_shear** definition calculates shear stress reduction in the lee of plants using the general method proposed by Okin (2008) for 2D simulations. If **process_okinR0var** is set to True, the R_0 coefficient is calculated using a quadratic relationship based on specific coefficients and tiller density for low wind speed (5.97 m/s) and high wind speed (9.58 m/s). The R_0 value is then interpolated to match the current wind speed. If **process_veglag** is True, the Okin method is applied at a distance behind the first vegetation, equivalent to the lag distance calculated in the **adjustment_length** definition. If **process_vegflex** is True, vegetation height is dynamically adjusted based on wind speed and tiller density, with calculations handled by the **VegFlexure** definition.

The **compute_raupach_shear** function calculates shear stress reduction in the lee of plants using the general method proposed by Raupach (1992) for 2D simulations. If **process_veglag** is True, the Raupach method is applied at a distance behind the first vegetation, corresponding to the lag distance from the **adjustment_length** definition. If **process_vegflex** is True, vegetation height is dynamically adjusted based on wind speed and tiller density, with computations handled by the **VegFlexure** definition.

For the 1D simulations, parameters for the previous processes (i.e., **process_veglag**, **process_vegflex**, and **process_okinR0var**) are directly calculated within the **vegsh ear_raupach** or **vegsh ear_okin** definitions, depending on whether they are True or False.

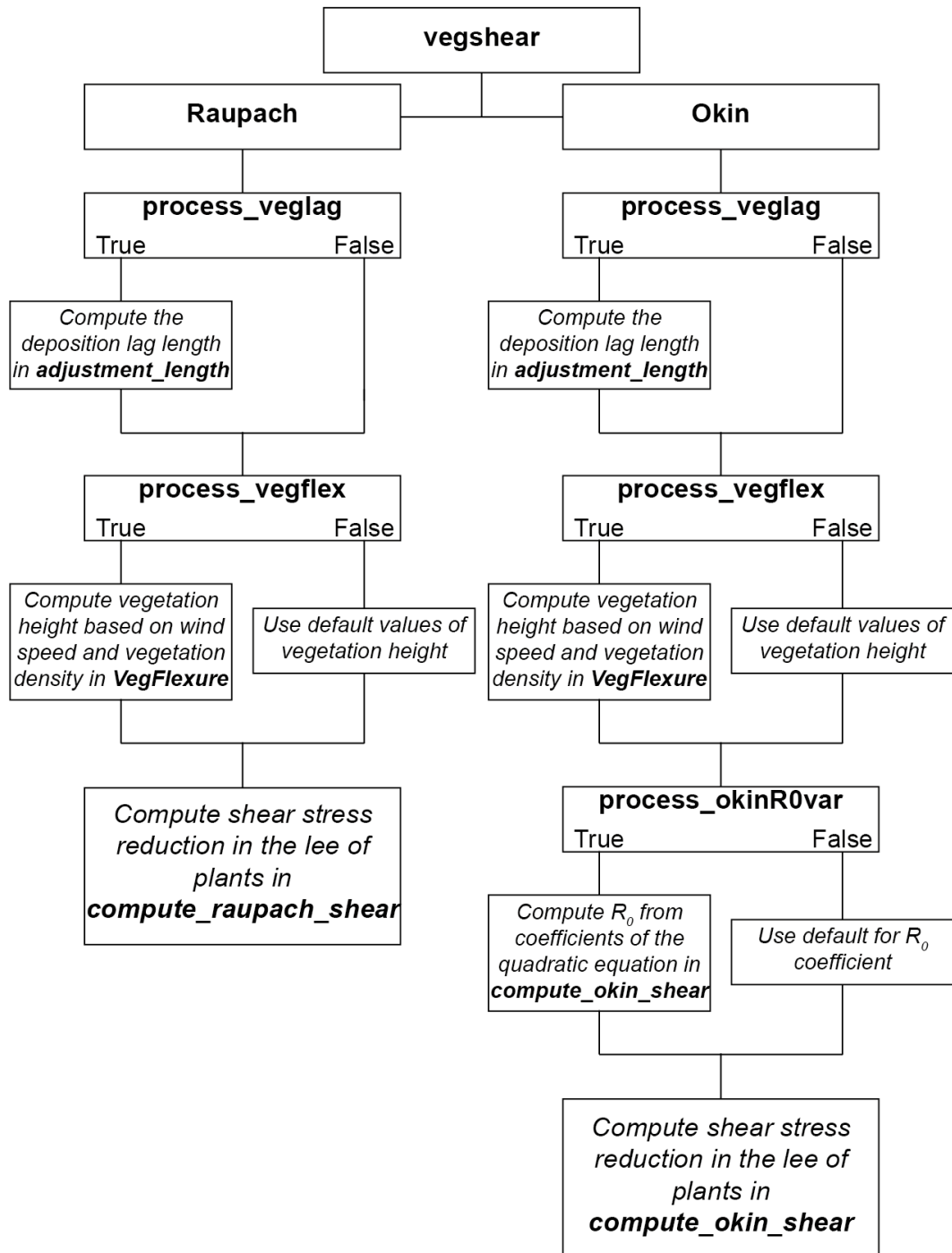


Figure 2. Flow chart illustrating how the new implementations are integrated into the **vegshear** definition for 2D simulations using the Raupach (1992) and Okin (2008) approaches.

4. References

Bradley, E.F., Mulhearn, P. J. (1983). Development of velocity and shear stress distributions in the wake of a porous shelter fence. *Journal of Wind Engineering and Industrial Aerodynamics*, 15(1–3), 145–156. [https://doi.org/10.1016/0167-6105\(83\)90185-X](https://doi.org/10.1016/0167-6105(83)90185-X)

- de Vries, S., Hallin, C., van IJzendoorn, C., van Westen, B., Cohn, N., Strypsteen, G., et al. (2023). AeoliS (version v2.1.0) [Software]. 4TU Repository. <https://doi.org/10.4121/22215562>
- Dickey, J., Wengrove, M., Cohn, N., Ruggiero, P., Hacker, S.D. (2023). Observations and modeling of shear stress reduction and sediment flux within sparse dune grass canopies on managed coastal dunes. *Earth Surface Processes and Landforms*, 48(5), 907–922. <https://doi.org/10.1002/esp.5526>
- Durán, O., Herrmann, H.J. (2006). Vegetation Against Dune Mobility. *Phys. Rev. Lett.*, 97, 188001. <https://doi.org/10.1103/PhysRevLett.97.188001>
- Hoonhout, B. M., de Vries, S. (2016). A process-based model for aeolian sediment transport and spatiotemporal varying sediment availability. *Journal of Geophysical Research: Earth Surface*, 121(8), 1555–1575. <https://doi.org/10.1002/2015JF003692>
- Laporte-Fauret, Q., Wengrove, M., Ruggiero, P., Hacker, S. D., Cohn, N., Zarnetske, P. L., Piercy, C. D. (2024). A new approach to account for species-specific sand capture by plants in an aeolian sediment transport and coastal dune building model. *Journal of Geophysical Research: Earth Surface*, 129, e2024JF007867. <https://doi.org/10.1029/2024JF007867>
- Okin, G.S (2008). A new model of wind erosion in the presence of vegetation. *Journal of Geophysical Research*, 113(F2), F02S10. <https://doi.org/10.1029/2007JF000758>
- Raupach, M.R. (1992). Drag and drag partition on rough surfaces. *Boundary-Layer Meteorology*, 60(4), 375–395. <https://doi.org/10.1007/BF00155203>
- Zarnetske, P.L., Hacker, S.D., Seabloom, E.W., Ruggiero, P., Killian, J.R., Maddux, T.B., Cox, D. (2012). Biophysical feedback mediates effects of invasive grasses on coastal dune shape. *Ecology* 93, 1439–1450. <https://doi.org/10.1890/11-1112.1>